Symbolic Projection Notes for $B_{c1} \to B_c^{(*)}\gamma$

Working notes

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1 Scope

These notes collect the compact FeynCalc projection results for the heavy-heavy channels

$$B_{c1}(6743) \to B_c\gamma, \qquad B_{c1}(6743) \to B_c^*\gamma, \qquad B_{c1}(6750) \to B_c\gamma, \qquad B_{c1}(6750) \to B_c^*\gamma.$$

The calculation is structurally different from the D_s and B_s light-cone sum rules. In the B_c system both valence constituents are heavy, so the leading radiative OPE is a hard-photon heavy-heavy calculation. There is no light spectator line to be replaced by photon distribution amplitudes, and no $\chi_s\langle\bar{s}s\rangle$ type term.

2 Currents and Propagators

For $B_c^+ \sim c\bar{b}$, we use

$$j_5 = \bar{b}\, i\gamma_5 c, \qquad j_\rho^V = \bar{b}\,\gamma_\rho c,$$

and the two axial basis currents

$$J_\mu^A = \bar{b}\,\gamma_\mu\gamma_5 c, \qquad J_\mu^B = \frac{i}{m_b + m_c}\,\bar{b}\,\sigma_{\mu\alpha}P^\alpha\gamma_5 c, \qquad P = p + q.$$

The physical currents are

$$J_{1\mu} = \sin\theta\, J_\mu^A + \cos\theta\, J_\mu^B, \qquad J_{2\mu} = \cos\theta\, J_\mu^A - \sin\theta\, J_\mu^B.$$

The free heavy-quark propagator in momentum space is

$$S_Q^{(0)}(k) = \frac{i(\not{k} + m_Q)}{k^2 - m_Q^2 + i0}, \qquad Q = c, b.$$

The hard photon is included by inserting the electromagnetic vertex on either heavy line:

$$S_Q^{(\gamma)}(k, k+q) = S_Q^{(0)}(k+q)\, ie_Q\gamma_\nu\epsilon^\nu\, S_Q^{(0)}(k).$$

For the $\bar{b}$ line it is convenient to use the effective antiquark charge $e_{\bar{b}} = +e/3$, while $e_c = +2e/3$. Equivalently, one may keep $e_b = -e/3$ and track the orientation of the antiquark line. The scripts use $e_{\bar{b}}$ explicitly.

Thus, yes: this is the same heavy-heavy propagator logic as in a B_c-mixing sum-rule calculation. The important difference from the strange-sector radiative calculation is that we do not insert a light-quark photon DA. If we later include power corrections, the natural heavy-heavy correction is the background-gluon expansion of the heavy propagators, e.g. gluon-condensate terms, not strange-quark condensate photon DAs.

3 $B_{c1} \to B_c\gamma$: E1 Projection

For the pseudoscalar final state we project onto

$$S_{\mu\nu} = p_\nu q_\mu - (p\cdot q) g_{\mu\nu}, \qquad \mathcal{P}_{\mu\nu} = \frac{S_{\mu\nu}}{2(p\cdot q)^2}.$$

FeynCalc gives the normalization check

$$S_{\mu\nu}\mathcal{P}^{\mu\nu} = 1.$$

We denote

$$p^2 = p_2, \qquad p\cdot q = pq.$$

After rewriting loop scalar products in terms of inverse denominators and setting those denominators to zero, the projected triangle-core numerators are

$$\begin{aligned} C_A^{(c)} &= -4i\,m_c + \frac{4i\,m_b m_c^2}{pq} - \frac{4i\,m_c^3}{pq}, \\ C_A^{(\bar b)} &= -4i\,m_b + \frac{4i\,m_b^3}{pq} - \frac{4i\,m_b^2 m_c}{pq}, \end{aligned}$$

so that

$$C_A = e_c C_A^{(c)} + e_{\bar b} C_A^{(\bar b)}.$$

For the tensor current,

$$\begin{aligned} C_B^{(c)} &= \frac{4i\,m_b m_c - 8i\,m_c^2}{m_b + m_c} - \frac{4i\,m_c^2 p_2}{(m_b+m_c)pq}, \\ C_B^{(\bar b)} &= -\frac{4i\,m_b m_c}{m_b + m_c} - \frac{4i\,m_b^2 p_2}{(m_b+m_c)pq}, \end{aligned}$$

and

$$C_B = e_c C_B^{(c)} + e_{\bar b} C_B^{(\bar b)}.$$

The denominator-cancellation terms are stored in the corresponding output file. They should not be discarded automatically; they correspond to reduced two-denominator or contact contributions and must be treated in the dispersion/Borel analysis.

4 $B_{c1} \to B_c^*\gamma$: Levi-Civita Projection

For the vector final state a useful gauge-invariant structure is

$$V_{\mu\nu\rho} = p_\nu\,\epsilon_{\mu\rho pq} - (p\cdot q)\,\epsilon_{\mu\nu\rho p}.$$

It is transverse in the photon index for $q^2 = 0$. The FeynCalc checks are

$$V_{\mu\nu\rho}V^{\mu\nu\rho} = -4p_2(pq)^2, \qquad V_{\mu\nu\rho}\mathcal{P}^{\mu\nu\rho} = 1.$$

The projected J_A triangle cores are

$$\begin{aligned} C_{A,V}^{(c)} &= -\frac{4i\,m_b m_c}{p_2} + \frac{2i\,m_c^2}{p_2} + \frac{4i\,m_c^2}{pq}, \\ C_{A,V}^{(\bar b)} &= -\frac{2i\,m_b^2}{p_2} + \frac{4i\,m_b m_c}{p_2} + \frac{4i\,m_b^2}{pq}. \end{aligned}$$

The tensor-current vector cores are

$$
\begin{aligned}
C^{(c)}_{B,V} &= \frac{2i\,m_c}{m_b+m_c} + \frac{2i\,m_b^2 m_c - 4i\,m_b m_c^2 + 2i\,m_c^3}{(m_b+m_c)p_2} \\
&\quad - \frac{4i\,m_b m_c^2 - 4i\,m_c^3}{(m_b+m_c)pq} + \frac{2i\,m_c\,pq}{(m_b+m_c)p_2}, \\
C^{(\bar b)}_{B,V} &= \frac{2i\,m_b}{m_b+m_c} + \frac{-6i\,m_b^3 + 4i\,m_b^2 m_c + 2i\,m_b m_c^2}{(m_b+m_c)p_2} \\
&\quad - \frac{4i\,m_b^3 - 4i\,m_b^2 m_c}{(m_b+m_c)pq} + \frac{2i\,m_b\,pq}{(m_b+m_c)p_2}.
\end{aligned}
$$

Again,

$$
C_{A,V} = e_c C^{(c)}_{A,V} + e_{\bar b} C^{(\bar b)}_{A,V}, \qquad C_{B,V} = e_c C^{(c)}_{B,V} + e_{\bar b} C^{(\bar b)}_{B,V}.
$$

5 Next Step

The double-dispersion variables satisfy

$$
P^2 = p^2 + 2p\cdot q, \qquad pq = \frac{P^2 - p^2}{2}.
$$

For $B_c\gamma$, the structure is close to the strange-sector E1 case, but with two heavy masses. For $B_c^*\gamma$, the additional $1/p_2$ and pq/p_2 structures mean that the double dispersion should be written explicitly before numerical Borel analysis. A diagonal single-variable replacement can be useful as a calibration check, but it should not be the final derivation for the vector final state.